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Exploring the potentials of more-than-human design for technology education

Gloria Felicia Lauterbach[1] · İdil Gaziulusoy[1] · Ricardo da Silva Torres[2,4] · Claudia López-Alfaro[2,3]

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Abstract
The shift from human-centric to more-than-human-centric is gaining attention across disciplines. Against the backdrop of the environmental crisis in general and biodiversity loss in particular, meaningful steps can be achieved through this paradigm shift. However, both outgrowing from long-held paradigms and growing into new worldviews takes time, even when the ever-accelerating environmental crisis requires urgent action. Interdisciplinary collaborations in education may help accelerate this shift. In this paper, we present a collaboration between Department of ICT and Natural Sciences, Immersive Technologies (NTNU) and Department of Design (Aalto University) as a case of how more-than-human perspectives can be incorporated into technology education within an existing curriculum. The aims of this paper are threefold. First, to highlight the importance of embedding more-than-human perspectives in technology education to challenge the prevailing human-centric focus in human-computer interaction. Second, based on current literature and the presented collaboration, to provide a methodological framework of how more-than-human perspectives can be embedded into technology education through three dimensions: cross-disciplinary collaborations, integration of values, knowledges and ethics based on reciprocity, and expansion of perspectives through immersion into renewed collaborations. Third, by presenting the case, to critically reflect on the practical implications of embedding more-than-human perspectives into technology education curricula.

Keywords Technology education · More-than-Human Design · Immersive technology · Sustainability transformations · Interdisciplinary education

✉ Gloria Felicia Lauterbach
gloria.lauterbach@aalto.fi

[1] NODUS Sustainable Design Research Group, Department of Design, Aalto University, Espoo, Finland

[2] Department of ICT and Natural Sciences, Norwegian University of Science and Technology Ålesund, Ålesund, Norway

[3] Departamento de Ciencias Ambientales y Recursos Naturales Renovables, Universidad de Chile, Santiago, Chile

[4] Artificial Intelligence Group, Wageningen University & Research, Wageningen, The Netherlands

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Introduction

To address urgent sustainability problems, we can no longer rely on incremental improvements in our socio-technical systems; there is a need for structural and societal transformations to achieve sustainable futures (Loorbach, 2010). These transformations not only call for technological development but also radical changes in institutions, organizations, and values/worldviews that underpin our current culture (O'Brien, 2018). As design researchers, sustainability scientists and technology researchers who think, act, and educate amidst polycrisis times and within educational institutions, we are confronted with the challenge of how to translate a more-than-human (MtH) worldview from thinking into action and through education. Contrary to a human-centric worldview, an MtH perspective makes way for the understanding that humans are in constant entanglement with and heavily dependent on the MtH world. Considering the fast-progressing decline of biological diversity on the planet, this conceptual shift is necessary and requires, before anything, a methodological shift in our approach to thinking, acting, and educating. In an article published over a decade ago, Krznaric (2012) proposed a revolution in human relationships as a means of addressing the global climate crisis. Krznaric identified empathy as a key mechanism for translating the vast knowledge that humans hold about the required actions into tangible ones. Empathy, therefore, functions both as an ethical and strategic guide for humans to confront the global crises we are facing. According to Krznaric (2012), empathy can be generated through three methods: educational learning, offering direct experiences, and enabling conversations between people. All three methods provide valuable coordinates when approaching the shift from an anthropocentric to a post-anthropocentric worldview through education. To think, act, and educate differently means to embark on open-minded collaborations; to test together empirically before knowing potential outcomes; radically changing methods; combining approaches; and engaging with students of entirely different knowledge domains by learning together. In this study, we explored how MtH perspectives can be included in technology education. This exploration is based on a small-scale course collaboration between the Department of ICT and Natural Sciences, Immersive Technologies (NTNU) and the Department of Design (Aalto University) and infused with the most recent practices of other scholars towards MtH educational framework building. Through our course collaboration, we aimed to answer the following research questions:

1) What key steps can be identified in theory when aiming to integrate MtH perspectives into technology educational curricula?
2) How do these key steps resonate in a practical case that aims to integrate MtH perspectives into technology education curricula?
3) How does MtH design help increase awareness of MtHs in developers of immersive technologies and what could be learned from the limitations observed in practice?

Exploring the potentials of MtH design for technology education: a preliminary framework

The starting point of this article and the research presented here is the need to move beyond the currently dominant human-centric approaches in general, and in the development of our built environments in particular. The main reasons for this starting point are threefold. First, human-centric approaches lack the means to consider and include the perspectives of those affected by human decisions (Forlano, 2017; Jones & Jones, 2020). Second, human-centricity ignores the irreversible dependencies of human society on the MtH world (Neumayer, 2012), which is typically amplified in technology development contexts, technology-developing disciplines, and technologically-optimist cultures of production and consumption (Oliveira Neto et al., 2018). Third, human-centric approaches fall short in engaging with urgent sustainability crisis, which requires attending to the complexities of socio-technical-ecological systems (Clarke et al., 2019; Ulmer, 2017), and in some instances require compromises of long-normalized human lifestyle privileges, such as (over)consumption of natural resources, particularly in economically advanced segments of society (Heikkinen et al., 2024). The discourse on MtH considerations regarding the built environment has existed for at least a couple of decades in the humanities and social sciences; however, the topic has only recently found resonance in literature that is situated in technical sciences. For example, recently there has been a call to adopt a MtH perspective in responsible innovation research so that stakeholder definitions can be expanded, and concepts of care and relationality can be embedded in technology development (Szymanski et al., 2021).There are similar echoes in research on technology education, with emerging yet increasing attention being paid to the inclusion of MtH perspectives in the engineering and design disciplines. While the attention is increasing, due to its emergent characteristic, the attempts to develop curricula that combine technology education and MtH considerations are currently limited. We provide an overview of these initial steps towards such a development, with the aim of advancing the field and by iteratively probing towards MtH educational frameworks.

Respectively, from selected background literature a set of requirements for the development of renewed technology education curricula are identified: Hansen et al. (2022) examine the inclusion of MtH in engineering education by elaborating how knowledge systems, knowledge acquisition, and knowledge exchange within research and higher education in the Western context can act as barriers to such endeavors. The authors call therefore for a technology education curricula in which knowledge systems are less hierarchical, knowledge originating from other cultures is considered with care so as not to decontextualize and reappropriate and cooperation between higher education institutions and industries in the development and application of new knowledge and technologies is a reciprocal and value-driven process. These coordinates illustrate the mindset shift that is needed not only by different stakeholders, such as researchers and industry, but also by those directly involved in learning environments. Accordingly, Bekker et al. (2023) approach MtH-inclusive curricula in human-computer interaction (HCI) education by identifying the challenges faced by teachers and students. After reviewing existing literature, they highlight representation,

inclusion, human and non-human designers, outcomes and effects, the role of technology, and bias as important aspects to consider when reworking curricula. The first four aspects have in common challenging questions about how to give voice to and include agents that have been muted or severely underrepresented. To represent MtHs, the authors argue, requires alternative forms of collaboration based on a broad ethical grounding. To include MtHs into human-centered, technological systems requires dealing with and accepting work of increased complexity and being open to new perceptions. To reposition *oneself* and MtHs in the process of making, requires working with renewed tools and methods to not take along human-centric perspective into the future. Understanding the value of renewed forms of collaboration by valuing their outcomes requires critically reflecting evaluation criteria about who benefits from a collaboration and what a successful collaboration means. The other aspects highlighted by the authors, the role of technology and bias, share the issues of critical engagement with technology. Using technology requires asking if, how and when technology can support the process of including and giving voice to other/s and when it simply cannot. The use of technological systems also requires a growing awareness of the contexts for which these systems have been developed, and how non-Western theories/practices are not easily assimilated into and connected with Western ideologies. Based on these considerations, Nilsson et al. (2024) offered an emerging pedagogical framework consisting of the three pillars of concepts, methods and practice. The target audience for their educational framework is technology design students, and the three pillars focus on thinking, performing and acting responsibly when incorporating the perspectives of the MtH world in future technologies. For this updated education, the authors offer a teaching repository of selected concepts (e.g. Latour's Actor-Network-Theory, pluriversal perspectives), productive methods (e.g. MtH personas, MtH ethnography) and explorative prompts (e.g. increase connectedness with nature in urban space) that support students and teachers in this shift of perspective and in overcoming the above-mentioned challenges. The positions as summarized above show that implementing MtH perspectives in technology education frameworks takes time, poses challenges, but also formulates requirements for different stakeholders. Arguments in support of such a development come from various scholars from the field of HCI with a growing focus on MtH design (Canina et al., 2021; Clarke, 2019; Giaccardi & Redström, 2020; Hansen et al., 2022; Ianniello et al., 2022; Irish & Romkey, 2024; Lönngren, 2019; Nijs et al., 2020; Powers et al., 2021).

To integrate MtH perspectives into technological systems the following three opportunities are identified: The first is to create learning opportunities to consider technology development in the context of highly complex socio-technical-ecological systems (Gutierrez-Bucheli et al., 2026; Ianniello et al., 2022; Lönngren, 2019; Powers et al., 2021; Sharma, 2025). Gutierrez-Bucheli et al. (2026) emphasize, for example, place-based learning and the value of exposing engineering students to diverse cultural and multispecies perspectives to foster more holistic relationships between engineers and the environments in which they work. A conceptual suggestion by Ianniello et al. (2022) is to dehumanize emerging technologies while, at the same time, supporting the inclusion of MtH voices for a more accurate perception of the diversity and complexity of systems. Powers et al. (2021) foster climate literacy in the education of engineering students due to the role they play as engineers within socio-technical systems. The second opportunity is to facilitate a shift in the focus of technology education away from a functionalist orientation towards deeply engaging, inclusive

and reciprocal technological systems (Giaccardi & Redström, Irish & Romkey, 2024; 2020; Nijs et al., 2020). Such technological systems will change, as argued by Giaccardi and Redström (2020), from systems we control to systems we "cultivate, nourish, and feed" (p.44). Nijs et al. (2020) connect therefore worlds through thinking-with and making-with in MtH speculative fabulation electronics street workshops as a supportive practice towards such learnings. The third opportunity in integrating MtH perspectives into technological systems is to broaden the narrow and human-centered perspectives that still dominate technological development and technology education (Canina et al., 2021; Clarke, 2019; Eriksson, 2024b; Hansen et al., 2022; Metcalfe, 2024). Canina et al. (2021) argued for the role of creativity in the post-anthropocentric era, which could be that of an "assemblage between human and non-human entities that produce hybrid knowledge and experiences" (p.139). In the context of smart city planning, Clarke et al. (2019) discuss voice multipliers, such as biosensors, as an example to better understand the environmental implications on diverse stakeholders in urban shared environments. Or, by building *umwelt apparatuses*, Metcalfe (2024) explores with students how simulated holistic perspectives do not provide a more precise understanding about species perception, but instead generate curiosity to imagine, engage with and design for different perceptual worlds.

Based on this background we present a preliminary framework for embedding MtH considerations in technology education. Table 1 summarises the key requirements that need to be fulfilled in developing MtH-considerate technology education curricula aligned with the key opportunities that exist to embed MtH considerations in technology development. From these we synthesise three key dimensions of MtH-considerate technology education practice. As shown in Fig. 1, these key dimensions are interrelated and could be implemented into technology education curricula as consecutive learning steps. We adopt these dimensions as a theoretical and analytical framework in presenting our case study and findings.

Methodology

In this paper, we explore how MtH considerations can be incorporated in technology education in an opportunistic manner; when there is not a substantial opportunity for curriculum development, and there is a need to work with existing structures and predetermined learning outcomes. We present a case study in which we collaborated to develop and test ideas to embed MtH considerations in technology education in the context of an existing course and report our findings and reflections on the set of approaches we used. Based on our findings, we presented a preliminary framework for course design to assist technology educators with similar intentions. The case study (Yin, 2014) was a suitable approach given the exploratory nature of our research (Eisenhardt, 1989). In this section we explain the background of the case study, the interventions we developed to embed in the "normal" running of the course and our data along with collection and analysis methods. The data were analysed qualitatively, and no coding software was used.

No ethical approval was required for the presented course collaboration. The students gave permission for the use of the produced material verbally.

Table 1 The three dimensions of MtH technology education frameworks

Key requirements for technology education	Key opportunities for technology development	Sources	Synthesised key dimensions of MtH technology education
- less hierarchical knowledge systems - alternative forms of collaboration with extensive ethical grounding - handling and accepting work with increased complexity and being open to perceiving anew	- to learn to develop for highly complex socio-technical-ecological systems	Hansen et al., 2022 Bekker et al., 2023 Ianniello et al., 2022 Lönngren, 2019 Powers et al., 2021 Irish & Romkey, 2024 Sharma, 2025	**Cross-disciplinary collaborations** **Integration of values, knowledges and ethics based on reciprocity** **Expansion of perspectives through immersion into renewed collaborations**
- knowledge originating from other cultures considered with care (to not decontextualize and/or reappropriate) - cooperation as a reciprocal and value-driven process	- to facilitate a shift from functionalist towards deeply engaging, inclusive and reciprocal technological systems	Hansen et al., 2022 Giaccardi & Redström, 2020 Nijs et al., 2020 Irish & Romkey, 2024 Gutierrez-Bucheli et al., 2026	
- to understand the value of renewed forms of collaboration by valuing their outcomes - working with renewed tools and methods to not take along human-centric perspective into the future - better balancing benefits and limits of technological systems	- to expand the often too narrow and human-centred perspectives still dominant in technology development	Bekker et al., 2023 Canina et al., 2021 Clarke, 2019 Hansen et al., 2022 Metcalfe, 2024 Eriksson et al. 2024b	

Background

The opportunity to embed MtH considerations in technology education has risen in the context of an interdisciplinary, international consortium project we are all part of. The project, NorDark, aims to develop an understanding of how to mediate human and wildlife needs in built environment design with a specific focus on urban lighting in green areas during Nordic dark season.[1] NorDark aims to integrate expertise from environmental psychology, urban ecology, and human physiology and sleep science to provide insights into lighting design research. Computer science is another discipline present in the project, specifically facilitating knowledge integration and pilot testing of lighting solutions through the use of digital simulated environments, such as a digital twin (Leplat et al., 2024). In addition, sustainable design and artistic research expertise also play a role in the integration of knowledge through stakeholder engagement, transdisciplinary research facilitation and development of a generic MtH design framework. The NorDark project places significant emphasis on generating transformative knowledge. This refers to information that aids society and its institutions in comprehending data, leading to a shift in our approach to the management of built and natural environments (Siebenhüner et al. 2018, Lang et al., 2012).

[1] https://www.nordark.org (As of March 2026).

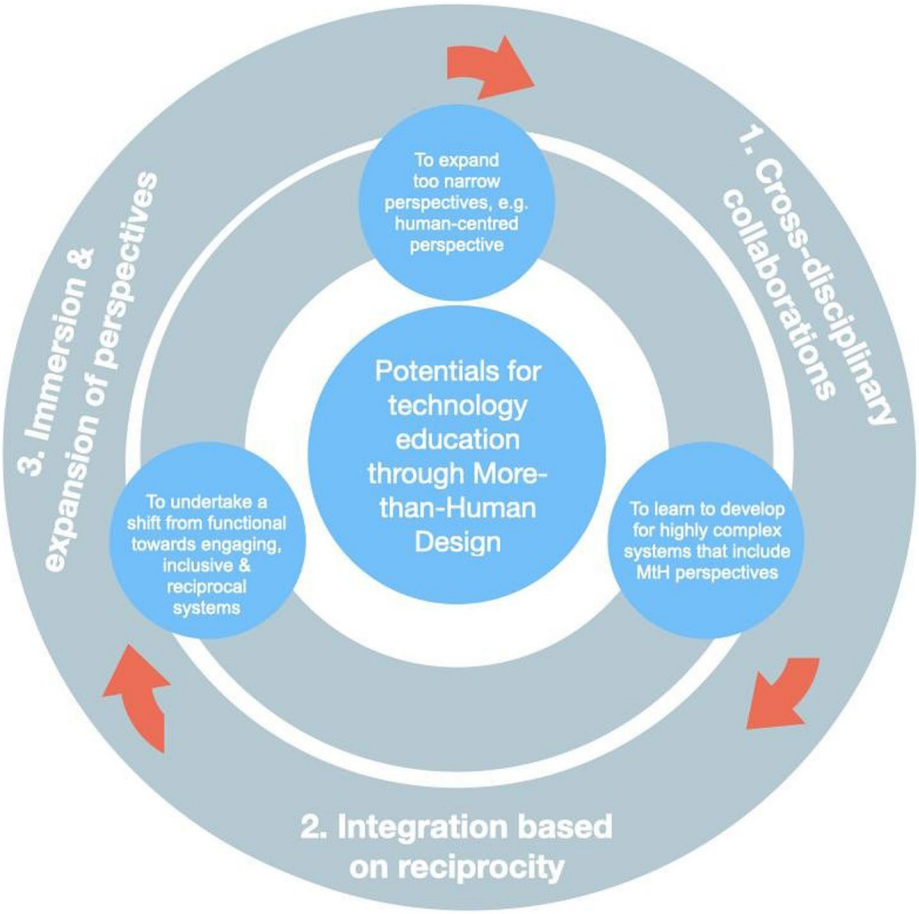


Fig. 1 Preliminary framework visualizing three dimensions and key opportunities relevant for implementing MtH design into technology education curricula

The collaboration for this case study took place between two of the five universities involved in the project and three of the seven work packages the project was organized through. Five researchers from this project were involved in this collaboration. The case study course, Immersive technologies, is part of the Simulation and Visualization Master of Science Program taught by the Department of ICT and Natural Sciences of the Norwegian University of Science and Technology (NTNU). Some of the classes were delivered online (e.g., theoretical lectures on concepts related to immersive technologies delivered by the course leader), while hands-on laboratory activities were conducted on campus. The researchers from Aalto University (AALTO) were responsible for introducing relevant concepts related to the MtH design topic and co-developed the project brief together with the course leader. An expert in ecology, who has been working across urban ecology and digital twin work packages, was a co-teacher of the course and assisted and guided students throughout the course (Fig 2).

The total number of enrolled students in the course was 11.

Embedding MtH considerations

The course began in January 2023 and ended in April 2023, spanning across 16 weeks. During those 16 weeks various course activities took place that dealt with the technical knowledge and skill development relevant to immersive technologies as well as MtH considerations that were specifically embedded into the course structure. The NorDark project was introduced to the students as part of the introductory lectures by the course leader as course context. The three assignments were cumulative and supported project-based learning through staging: the first assignment asking for a literature review on immersive technologies, the second assignment asked for the development of a prototype immersive environment and the third for development of the overall immersive experience of the prototype. The students were asked to develop a digital model of the NorDark project site in the city of Ålesund including lighting infrastructure as a modular component of the digital model (the locations and intensities of light could be altered to test different scenarios). The immersive experience of the prototype would be developed from the perspective of a local species students choose and for which vision and light response were available. The students had the option to work in groups or individually.

AALTO researchers were responsible for developing a program of activities to sensitise the students to become familiar with adopting a MtH perspective in technology development. Given that the primary learning objectives of the course were to develop technical knowledge and skills as required by their specialization, time effective ways needed to be found to equip the students with the necessary theories related to MtH design and to help them find the necessary information and data to use in their project. Below is the summary of these activities:

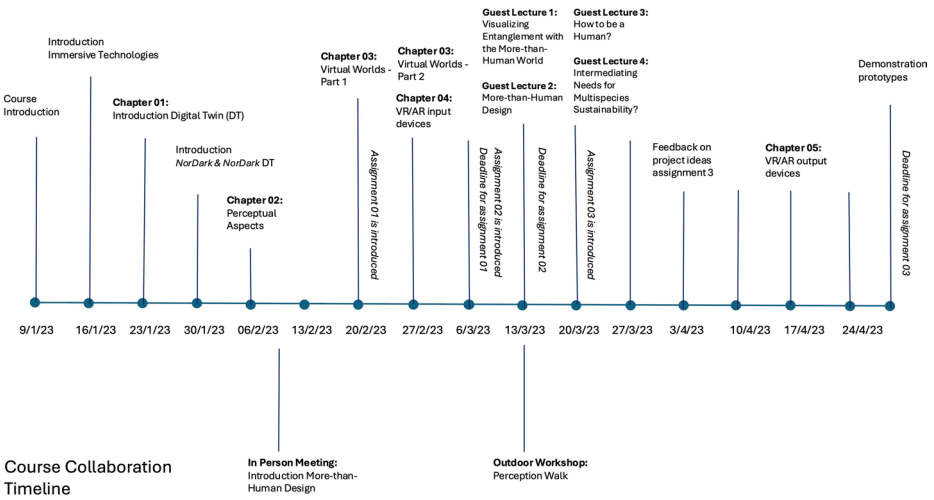


Fig. 2 Course collaboration timeline with relevant events associated with both theoretical and practical activities

In-person meeting with AALTO researchers

The students met with AALTO researchers in person in February 2023, taking the opportunity for a project consortium meeting on site. This was considered an initiation activity and mainly aimed at building rapport. The researchers introduced themselves, explained the broader aims of the NorDark project and set the general contextual scene of the course project with references to MtH considerations. They discussed with the students how they have understood the course assignment and how an immersive technology prototype that considers a MtH perspective could look like in practice. The meeting lasted about an hour.

Guest lectures by AALTO researchers

Four guest lectures were delivered online by AALTO researchers, two each over two consecutive weeks in March 2023. In the lectures, MtH design was explored through different theoretical lenses and examples from practice. After each lecture session the students had time to discuss in groups their learnings from each lecture.

Perception walk

This activity aimed to immerse the students in a real-life setting so that they would become sensitizied about the MtH qualities of the human built environment, learn to pay attention to and perceive MtH encounters. The walk was designed by AALTO researchers and conducted by the ecologist co-teacher of the course. It took place on March 20, 2023, on the grounds of the Sunnmøre Museum, located 15 min walking distance from the campus. The museum grounds feature historical houses as well as a water bay which is a habitat for aquatic bird species. In this activity, the students would work individually and quietly. The activity consisted of three parts and a final meeting to share the general insights. In each step, the students would walk through a path and at a self-selected spot would observe the area for at least five minutes and write a short note responding to the following questions: What do you see? Hear? Feel? Smell? Taste?. The first spot would be an area of 10×10 m; the second spot would be an area of 1×1 m, and the third spot would be an area of 0.1×0.1 m. The students would walk for about ten minutes in between selecting their next spot. The walking activity lasted for approximately 60 min, and the insight-sharing meeting was 30 min in total.

Expert help

During prototyping, students were given support to find and use resources about the vision and light sensitivity of the species they chose for their project by the course co-teacher ecologist. In addition, students had the possibility to discuss their questions and ideas with the ecologist co-teacher of the course. In addition, the AALTO researchers participated in the interim project presentation and gave formative feedback about the prototypes in regard to MtH considerations (Fig 3).

Data

The data we collected for our study include the following:

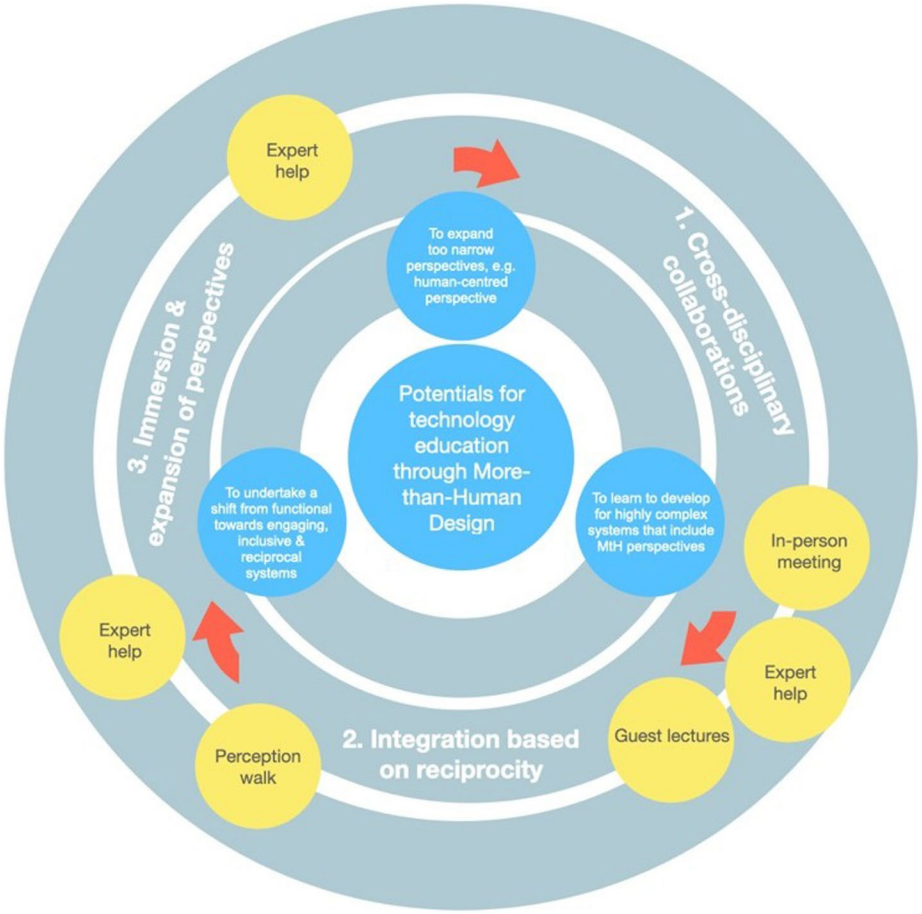


Fig. 3 Course activities mapped onto the preliminary framework

Observations

Each of us involved in the course had the opportunity to make observations at different points. For the two AALTO researchers the key moment of observation occurred during the in-person meeting with the students. The course leader and the co-teacher carried out observations throughout the course. The co-teacher was also the only one who was present during the perception walk. She was also present to the students during discussions following the guest lectures and in one-to-one meetings with the groups during prototype development regarding the students' challenges and questions in embedding MtH perspectives. Our observations complement the hard data collected and helped us to generate deeper insights about student experience of the course. We regularly shared and discussed our observations both during the course and during data analysis and writing of this article.

Notes from the perception walk

Seven out of 11 students participated in the perception walk. As instructed, they have responded to four perception questions in the form of short notes in three different locations during the walk. The 30-minute-long group discussion at the end of the walk was facilitated by the co-teacher of the course who ran the perception walk. Data, in addition to students' notes, also includes her notes about the activity and the group discussion. The notes from the students and the co-teacher were analysed qualitatively.

Prototypes

A total of six prototypes were produced by the students. Two students worked independently, one group consisted of three students and three groups were formed by two students each. Table 2 shows the group codes, number of students for each prototype, the selected species and the MtH objective pursued by the students. The prototypes provided visual data and complemented the analysis of the student reports. Figure 4 provides screenshots from each prototype.

Student reports

The final prototypes were accompanied by project reports. The students were asked to describe their objectives for the prototypes, findings in relation to the light-related behavior of the species they focused on, their prototype development process, the challenges they encountered during the project and future research questions stemming from their work. In addition, questions specifically formulated to understand the experience of students in working with MtH themes in immersive technology development were included in the assignment brief. These questions asked students to provide an overview of how the MtH design perspective was handled in the project, the challenges they faced specifically in relation to thinking from a MtH perspective and how the MtH theme of the project changed their overall way of working with immersive technology. There was a total of six reports corresponding to the six project groups.

Findings and discussion

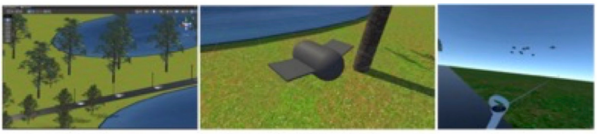
Cross-disciplinary collaborations

Among all the data sets, our observations and the student reports were the most insightful in understanding the extent of the course activities. As explained in the previous chapter, the arranged course activities were helpful in embedding MtH considerations into the overall learning of the students in the course. We tried to facilitate cross-disciplinary learning from the beginning of the course through three main mechanisms: in-person meeting of two AALTO researchers with the students, guest lectures delivered by AALTO researchers, and the availability of an ecologist as a co-teacher throughout the course. The in-person meeting revealed significant epistemological differences between the technology education formation of the students and the MtH design perspective of the AALTO researchers. The students

Table 2 List of prototypes developed by groups G1-G6

Group code	No. of students in the group	Focus species	Description
G1	2	Lesser horseshoe bat (*Rhinolophus hipposideros*)	This prototype aims to communicate the effect of streetlights on commuting bats. The behaviour of virtual bats is changed in response to the presence or absence of streetlights in the virtual environment.
G2	2	Mountain hare (*Lepus timidus*)	This prototype focuses on sudden behavioural changes of rabbits, conditioned to the presence of artificial lights. The simulation considers the perspective views of rabbits and external observers.
G3	2	Insects (unspecified)	This prototype focuses on simulating the effect of artificial light on insects. Users of the immersive experience can observe changes in the behaviour of insects subject to the presence of artificial light and food. The tools allow experiencing the insect view perspective.
G4	1	Human, bat (unspecified)	This prototype supports the experience of two perspectives: from humans (while jogging) and from bats. The design of the virtual light infrastructure considers both perspectives including the design of the spacing, the intensity, the color, and the orientation of light poles.
G5	1	Frog (unspecified)	This prototype aims to simulate the behavioural changes of frogs conditioned by artificial light. The focus is on simulating light variations in an urban wetland.
G6	3	Squirrel (unspecified)	The goal is to support the understanding of the effects of artificial light from the perspective of small arboreal animals through a gamified simulator. Users can experience the look from the animal's perspective, accounting for an increased field of view and the use of dichromatic colors.

had difficulty understanding what is meant by integrating MtH perspectives into a digital technology as they were unfamiliar with the discourses on decentring humans in design, care ethics and building empathy with MtHs. The lack of a common discourse as well as common terminologies made it difficult to establish a common ground. The students were mostly interested in receiving "data" from us to use in their simulation modeling, however, AALTO researchers put an emphasis on the conceptual shift required in developing simulations that embodied MtH perspectives. While this initial meeting was not productive in opening up and shifting the perspectives of the students, it was instrumental to build rapport and trust, make a point that "data" can be understood differently in different disciplines and may also come in different forms (quantitative versus qualitative). The meeting was also helpful for AALTO researchers to get acquainted with the epistemological perspectives of the students and enabled careful thinking about how to formulate the project brief together with the course teachers so that the language and terminology used were understandable and the project was workable by the students. These findings show that the cross-disciplinary approach is very relevant as a first step towards the MtH framing of technology education. As synthesized through the selected background literature and observed in this case, cross-disciplinary collaborations enable disciplines to come closer to one another by reducing the hierarchies between diverse knowledge holders (Bekker et al., 2023; Hansen et al., 2022). While AALTO researchers gave access to concepts and approaches from their fields, NTNU students revealed difficulties in understanding how to implement them on their own. Better



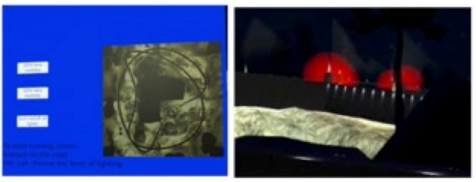
a) Prototype G1: virtual environment with lake, trees, streetlights and moving bats (left); 3D model of a bat (middle); illustration of the teleportation technique (right).



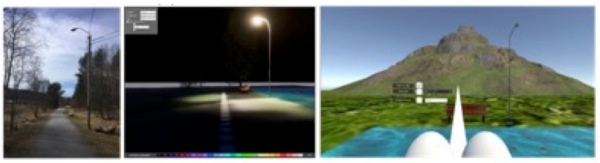
b) Prototype G2: virtual environment with vegetation and light poles (left); rabbit's view (middle); observer's view (right).



c) Prototype G3: virtual environment with light poles (left); human perspective (middle); insect perspective (right).



d) Prototype G4: a screen to change the light intensity in the simulation (left); examples of red lights used in the simulation (right).



e) Prototype G5: real-world area targeted in the NorDark project (left); virtual scene that mimics the real-world area (middle); virtual scene with a park terrain, some trees, a path, a pond, a streetlamp, a bench, and several frogs in the water (right).



f) Prototype G6: virtual environment with vegetation and light poles (left); a scene with dichromatic vision in daytime (middle); a scene with dichromatic vision with artificial light at nighttime (right).

Fig. 4 Representative screenshots from the prototypes

consideration of MtHs in technology development proved in this course collaboration to work as a shared goal, which motivated the whole group to think beyond their own disciplinary knowledge and language realms. For example, following the guest lectures, the students did not ask any questions except for in one case by one student. While this is not on its own an indication of disinterest or disengagement, during the discussions facilitated by the co-teacher following the guest lectures, the students expressed difficulty in understanding the importance and relevance of the guest lectures for their project. However, a few students asked questions about the lectures after some encouragement by the teachers and started pondering for the first time how MtH perspectives could be integrated into or visualized through digital technologies. Another valuable observation that can be made here is that in the student reports that were delivered at the end of the course, only one group (G3) mentioned the guest lectures as beneficial for the development of their prototype or for the overall course learnings. The results raise questions for the course teachers about the benefits of the guest lectures, the didactics used, and the order of the embedded activities. In other words, the findings indicate that cross-disciplinary collaborations require good preparation and a greater emphasis on teaching formats supporting the coming closer of different knowledge domains in order to facilitate long-lasting impact on students' learning outcomes. This is particularly true for MtH concepts, which require more than a few guest lectures to be effectively integrated into the curriculum.

Integration of values, knowledges and ethics based on reciprocity

Based on the data collected, the perception walk, an activity encouraging immersive engagement with the environment across different scales and engaging multiple senses, seemingly created a breakthrough for the students. While the beginning of the activity did not imply such development, the student notes taken during the activity suggest changes in perception towards MtHs. Similarly, as with the guest lectures, at the beginning of the perception walk activity, some students found it difficult to understand the purpose and showed resistance or a lack of enthusiasm. For example, some expressed that they could not understand how this activity could help them progress in the course and/or how they could acquire knowledge. This is perhaps indicative of how technology education is mostly conducted; indoors, often with computer aid, in classroom environments.

The notes suggest gradually increasing immersion, attunement and awareness (as the scale of focus gets smaller) regarding the presence of MtH elements in the green urban area site. For example, one student simply noted "*Wet rocks*, *spring*, *birds*, *melting snow*, *grass trough snow*, *pleople [sic] walking*, *sun*" in response to the seeing related question in the largest focus area (10 × 10 m), whereas for the same question in the smallest focus area and the third iteration in the activity (0.1x,0.1m), the student noted "*Dead leaves of the trees on the ground. Vey [sic] small plants with delicate lines and features. Green plants […] growing in the cold winter. Very delicate remains of a leaf that is decomposing. There are more than 50 different colors*, *very small branches with many details*". The notes also point to the perception of interactions among natural and built environments across different scales. For example one student captured seeing in the smallest scale "Shoe prints […] pushing into the snow" and for hearing "Water dripping from the roof of a building; can hear the drop hitting the snow". It was also remarkable that the students used terms and concepts not commonly employed in typical teaching and learning activities involving technology education with

references to not only animate but also inanimate elements. As an example, one student (G5) writes down to hear within the largest focus area: "Raven crying, driping [sic.] water from houses roof into lake and duck; One old house nearby […] electrmagnetic [sic.] noise from inside". Such terms/concepts are relevant in the design of immersive experiences that communicate MtH perspectives in urban green areas and use of which by the students indicates that the activity was also helpful in prompting students to start using language beyond technological terms in relation to their projects.

Despite the initial low motivation, it was remarkable that during the group discussion at the end of the activity, all students actively participated and discussed ideas for their prototypes on the basis of the insights they gained through the perception walk activity. All in all, the perception walk activity outdoors can be mentioned as a shifting experience. The students have developed a first understanding of post-anthropocentric values by showing interest and empathy towards MtHs. One student (G1) noted seeing in the third iteration (0.1x,0.1 m) "Precision of making order of narture [sic.]. Leaves holding on insect, maybe it is feeding the insect. The insect position is influenced by sunlight, I think". This note may point to an understanding of reciprocity that has surfaced through the exercise. The in-person meeting and guest lectures in the beginning of the course did not enable this step. The students started to connect scales of perception by thinking-with and making-with their environments, as discussed above, as a clear opportunity when integrating MtHs into technology development (Nijs et al., 2020). Surprisingly, despite student engagement with and learning in this activity being evident in students' field notes and our observations, none of the groups mentioned the activity in their reports except for one (G5). Again, as with the guest lectures, this result is a motivation for the teachers of the course to reflect on the actual impact of the course collaboration and how such meaningful and seemingly valuable activity can be further strengthened.

Expansion of perspectives through immersion into renewed collaborations

The ecology-related expert help made available to the students throughout the course, in addition to the expert help on immersive technologies, proved to be instrumental in enabling the students to embed MtH considerations into their prototypes. The students made ample use of this opportunity and discussed with the experts (mainly the two co-teachers of the course) their ideas and challenges in relation to building immersive experiences into their simulations using a MtH perspective. Most of the groups defined the scope of their prototypes at the early stages of the project implementation and conducted thorough investigation of their chosen species, however, they found it quite challenging to handle the resulting additional large amount of data and integrate it into the urban built environment related data consistently and meaningfully in building their prototypes. Through regular discussions with the experts, they were guided to clarify the focus of their project, the angle they will embed the MtH perspective into their simulations, and frame the scope of their immersive experience design in line with the time available for the project, thereby, thereby enabling them to learn how to work with complex and multi-layered data sets in feasible ways.

As discussed in context of our preliminary framework, bringing knowledge and language from different knowledge domains together, in our case MtH design and engineering, enables a step-by-step increase in topical complexity, which better represents the socio-environmental-technological systems in which we are embedded in (Ianniello et al., 2022;

Lönngren, 2019; Powers et al., 2021). In the course, we observed that the students were very engaged in learning about their species of choice and, while they found it challenging, they were also motivated to learn about the dynamic and complex nature of ecological processes and how to integrate ecological considerations into technology. The students also used scientific literature related to the light behavior of their chosen species, however, audio-visual media providing scientific information in easily understandable formats was found to be the most useful.

While integrating an MtH perspective into immersive technology was the common, high-level task set by the project brief, each group approached this task in slightly varying ways which is evident from the reports. The main motivations mentioned include: to help understand and tackle harmful impacts of artificial light on chosen species (G1, G6), to explore less harmful human cohabitation with chosen species by enabling a comparison between human vision and species vision in the dark (G4), to help understand the impact of artificial light on species behavior (G2, G5), and to observe and understand the movement and behavior of chosen species during night and day (G3). The impact of artificial lighting on the behavior of the species and its habitat was addressed in a prototype developed by G5.

These motivations also imply how the groups intended their simulations to be used, or for what purposes and for whom the simulations were useful. With the prototypes of G1, G2, G3, G5, and G6, users can gain insights into the behavior of a certain species through a visualization/simulation of the effects of artificial light on the chosen species. However, for G4's prototype, the user can experience the interaction between humans and MtHs by visualizing and simulating the effects of artificial light on both. The intended target users of the prototypes were quite diverse. All student groups aimed at identifying one or more user groups for their prototypes: lighting designers (G1), urban planners (G2, G3, G6), interested public (G6), policymakers (G4, G5), and educators (G3).

Future research projects identified by the groups in the student reports were also insightful in regard to understanding the extent to which the course activities stimulated the students to embed MtH considerations into their future work or see the relevance of this for other future projects. Except for one group (G4) which identified a solely human-centric future project (investigating the effect of different lighting conditions on human safety) all groups proposed future projects that either expanded the focus from a single species perspective (G1, G2, G3, and G5) or deepened their analysis by investigating the effect of different lighting conditions on the chosen species (G1, G3, G5, and G6). Among the future projects proposed by these groups include a comparative study on commuting behavior of multiple bat species exposed to different street light sources (G1), further investigating the behavior and stress levels of a local hare species exposed to human activities (G2), investigating the dynamic interactions of various species and the built environment under changing lighting conditions (G3), and further research on the social behavior and physiology of squirrels affected by the time period, color, and intensity of artificial light (G6). These future project ideas show the leap in learning the students made from the perspective of an engineer with a human-centric, functional perspective towards an engineer with an engaging, inclusive and reciprocal perspective. At the end of the course, the students were motivated to integrate into their future projects with new values and knowledges that acknowledged the role of MtHs within our complex socio-techno-ecological systems.

While we did not observe a major perspective change in students relating to MtHs, we did see renewed forms of collaborations the students engaged in because of the experimental

course structure. The students widened their perspectives by reviewing theories and concepts from MtH design and by reviewing literature, such as biology and ecology. Through the in-person meeting with AALTO researchers, guest lectures provided by AALTO researchers, the expert help and the perception exercises, the students engaged in thinking and acting on a different basis than they were used to. The technological systems they learned to develop were discussed and visualized in a context and as part of a socio-techno-environmental system. The immersive technologies became a tool in support of sustainability transformation compared to being a tool in support of further advancing technological systems. In addition, future projects suggested by the students in their course reports demonstrate the interest of most groups in collaborations beyond human-centricity. It is unlikely that these short-term collaborations will expand the perspectives of the involved students in the long run, yet such temporary shifts in perspective can nevertheless leave a mark when engaging in future, human-centric projects.

From the point of view of the involved researchers, both AALTO researchers and NTNU researchers, more preparation time would have been required to bring very different epistemological perspectives together and to better learn from each other's expertise. In the same regard, more in-person meetings would have been a more efficient way to find common grounds instead of stirring complex discussions online without any fruitful discussion. The researchers further learned that introducing MtH perspectives to engineering students requires far more preparation work tailored to their respective learning levels. Despite the ad-hoc circumstances of preparing the students for the course collaboration, the researchers experienced an increase in motivation by the students due to working with a real-life, complex case with direct benefits for society, wildlife, and nature. Especially for the ecology expert, it has been inspiring to see that the students developed empathy for MtHs in a very short amount of time. Repeating the course would have been the wish of all researchers as it could have benefitted the development of the preliminary framework through iteration. Repeating the course collaboration could also have helped develop the skills of the involved researchers for the benefit of the following student cohort. However, a second round was not possible. Nevertheless, both AALTO and NTNU staff reported personal learning experiences because of the transdisciplinary nature of the case. Another observation from the point of view of the involved researchers was that considering that course scripts are normally fixed and rigid, the students dealt with the experimental set-up very well. Bringing MtH considerations onto the table of engineering students who have never worked from an MtH-centric perspective has been pedagogically successful: the students not only learned technical concepts and skills in the course but also had to place and test the newly learned skills in an entirely unknown territory.

Conclusions

In this article, we presented a course collaboration organised between AALTO and NTNU , which enabled us to explore how MtH perspectives can be integrated into technology education. Based on the aforementioned course collaboration, we further developed and discussed a preliminary methodological framework comprising three key dimensions for integrating MtH perspectives step-by-step into technology education curricula. The outcomes of the presented course collaboration – the MtH considerate immersive technology prototypes – have been developed in support of the activities AALTO researchers have integrated into

the already existing curriculum structure of the course. The key dimensions of the preliminary framework were synthesized through background literature of the key requirements and opportunities for technology educational practice when integrating MtH perspectives. The three key dimensions are (1) cross-disciplinary collaborations, (2) integration of values, knowledges and ethics based on reciprocity, and (3) expansion of perspectives through immersion into renewed collaborations.

The approach we have chosen based on what was possible in the given context – the collaboration's ad-hoc and short-term nature – marks both its limitations and its potentials. The collaboration would have benefited from a longer, more intensive preparation phase, an extended course duration and opportunities to iterate and adjust the format. Given the multiple crises of our time, the comprehensive changes necessary in technology education curricula should be informed by in-depth, iteratively developed, and comparable data but also generated by many and across diverse, experimental settings. Accordingly, the framework we propose relies on combining the experiences, requirements and opportunities as researched by others with the experimental experiences and outcomes through practice we have generated. With our framework proposal, we wish to contribute to a renewed engineering education curricula that teaches future technology students ways to participate in our complex socio-technical-ecological systems rather than just ways of making and unmaking them according to humans needs. In our case, the involvement of an ecology expert and the outdoor activity were most beneficial aspects of the collaboration in that regard—likely to still inspire the involved students over time. For us as researchers and authors, the students' clear progress because of this different approach—nurturing their curiosity about an unfamiliar field with expert guidance and teaching engineering students outdoors—was especially evident. Here, we see future research opportunities but also clear and direct possibilities for low-threshold implementation into the educational practice in the field of engineering through design research and design education.

As Eriksson et al. (2024a) presented in their scoping review, design research is at the forefront of MtH-centric education. In addition, the same authors argued that environmental sustainability appears to be the most important motivation for such shift in worldview. In light of the fact that our built environments are, for the most part, of a technological nature, it seems clear that collaboration between design research and technology disciplines is of great value. This is because the insights and skills of design researchers can be lifted into the disciplines that engineer the systems and infrastructures in need of sustainability transformation. Learning how to consider and include MtHs and building awareness for MtHs is a key aspect in the work of MtH design. On the other hand, technologies, such as Virtual Reality and Augmented Reality, are valuable tools that make a change in perspectives possible. Yet, often, those who develop tools that are used for making informed decisions about our built environments are being educated based on human-centered values and develop technology based on "what is possible [instead of] what is responsible for the planetary well-being" (Bekker et al., 2023, p. 56). A sustainable transformation of our built environments requires developers of immersive technologies to expand their definitions and perceptions of who all is affected by our built environments and embed concepts of care into the development of immersive technologies (Szymanski et al., 2021).

Our case collaboration aimed to inspire, advance, and methodologically ground MtH approaches within technology education by offering practical guidance.

The proposed framework as developed in this paper can support future collaborations to navigate challenges of transdisciplinary course curricula on the crossroads of engineering and design education.

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Declarations

Conflict of interest No potential conflict of interest was reported by the authors.

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